

1. Administrative & Study Identification

Full Study Title: A Study to Assess Drug Levels, Tolerability and Absolute Biological Availability of Single-dose of BMS-986446 in Healthy Participants **Short Title/Acronym:** BMS-986446 SC vs. IV Bioavailability Study **Protocol Number:** CN008-0026 **NCT Number:** NCT06955741 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** BMS-986446 (formerly PRX005, Anti-MTBR Tau Antibody) **Phase of Development:** Phase 1 **Medical Monitor:** Bristol Myers Squibb Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess drug levels, tolerability, and the absolute biological availability of a single dose of BMS-986446 administered via subcutaneous (SC) versus intravenous (IV) routes in healthy participants. **Secondary Objectives:** To evaluate the safety, pharmacokinetics, and injection site tolerability associated with the SC administration of BMS-986446.

2.2 Background & Rationale BMS-986446 is a potential best-in-class antibody under development for the treatment of Alzheimer's disease. It specifically targets a key epitope within the microtubule-binding region (MTBR) of the tau protein. This Phase 1 study is necessary to determine the absolute bioavailability and tolerability of subcutaneous administration compared to standard intravenous administration, supporting the development of a more convenient, less invasive dosing option for patients in future clinical trials.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (IV vs. SC administration) **Blinding:** None (Open-label) **Randomization:** Randomized (Parallel Assignment)

3.2 Planned Enrollment & Duration Target \$N\$: 46 **Trial Status:** COMPLETED **Number of Sites:** Multicenter Clinical Pharmacology Units **Estimated Study Start:** 2024

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Participants must have a Body Mass Index (BMI) of 18.0 to 30.0 kg/m2. 2. Participants must be

healthy as determined by medical history, physical examination, neurological examination, vital signs, 12-lead ECG, C-SSRS, and clinical laboratory evaluations. 3. Ability to understand the purpose and risks of the study and provide signed informed consent.

4.2 Exclusion Criteria 1. Participants must not have clinically significant gastrointestinal, renal, hepatic, broncho-pulmonary, neurological, psychiatric, cardiovascular, endocrinological, immune-mediated disorder, hematological, ongoing allergic disorder requiring treatment, metabolic disorder, cancer, or cirrhosis. 2. Participants must not have prior exposure to BMS-986446 (PRX005). 3. Other protocol-defined inclusion/exclusion criteria apply, including inability to comply with study constraints.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Single-dose administration. **Route of Administration:** Subcutaneous (SC) injection versus Intravenous (IV) infusion. **Duration of Treatment:** Single administration on Day 1, with safety and pharmacokinetic follow-up to approximately Day 85.

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications approved by the investigator for healthy volunteers (e.g., occasional paracetamol for mild ailments). **Prohibited:** Use of any other investigational drug or biological product; medications that could interfere with pharmacokinetic assessments.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, BMI check, ECG, Neurological & Lab Assessments
Baseline	Day 1	Randomization, Single Dose Administration (SC or IV), Intensive PK sampling
Interim Follow-up	Days 2 to 84	PK blood draws, Injection site reaction questionnaires, Vital Signs, Safety Labs
End of Study	Week 12 (Day 85)	Final PK sampling, Exit Interview, Final Safety Assessment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: 1. Geometric Mean Ratio (GMR) of Maximum Observed Serum Concentration (C_{max}) of BMS-986446 Administered IV versus SC. 2. GMR of Area Under the Serum Concentration-time Curve From Time Zero to Time of Last Quantifiable Concentration (AUC(0-T)) of BMS-986446 Administered IV versus SC. 3. GMR of Area Under the Serum Concentration-time

Curve From Time Zero to 672 Hours (AUC(0-672)) of BMS-986446 Administered IV versus SC. **Measurement Method:** Serial pharmacokinetic blood sampling followed by central bioanalytical laboratory analysis.

7.2 Secondary & Exploratory Endpoints 1. Number of Participants With Adverse Events (AEs). 2. Number of Participants With Serious Adverse Events (SAEs). 3. Number of Participants With AEs Reported Related to BMS-986446. 4. Tolerability of SC injection, assessed using participant-reported questionnaires rating pain (Numeric Rating Scale), itching, burning, pressure, and soreness/tenderness at the injection site.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard Phase 1 intensive monitoring; participant-reported response for injection site tolerability using specific rating scale questionnaires. **Serious Adverse Events (SAEs):** Evaluated continuously and reported within standard 24-hour safety regulatory timelines. **Data Monitoring Committee (DMC):** Internal safety review committee overseeing cohort progression and data integrity.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Pharmacokinetic (PK) Set (for C_{max} and AUC parameters) and Safety Set (for all dosed participants). **Sample Size Justification:** Powered adequately to estimate absolute bioavailability and define the 90% confidence intervals for the C_{max} and AUC ratios between SC and IV administration. **Handling of Missing Data:** Missing PK samples generally not imputed; standard non-compartmental analysis (NCA) applied to available serum concentration data.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Rigorous onsite monitoring of source data, focusing on informed consent, drug accountability, PK sampling timestamps, and AE reporting. **Audit Trail:** eCRF locking, automated PK sample tracking, and data clarification forms handled through standard data management procedures prior to database lock.

11. Study Identifiers & Governance

Posting ID: 714534705788289739 **Primary ID:** CN008-0026 **Registry Number:** NCT06955741 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Assess Drug Levels, Tolerability and Absolute Biological Availability of Single-dose of BMS-986446 in Healthy Participants **Official Regulatory Title:** A Phase 1, Randomized, Open-label, Parallel, Single-dose Study to Assess the Pharmacokinetics, Tolerability, and Absolute Bioavailability of Subcutaneous Administration of BMS-986446, an Anti-MTBR Tau Monoclonal Antibody, in Healthy Participants

12. Clinical Framework (Study Specific)

Primary Condition: Healthy Volunteer (Evaluation for Alzheimer's Disease candidate) **Diagnosis Threshold:** Healthy status confirmed by BMI (18.0 to 30.0 kg/m²) and normal clinical exams **Medical Methodology:** Pharmacokinetic (PK) and Bioavailability Assessment **Optimization Target:** Subcutaneous (SC) route feasibility and bridging

13. Study Procedures & Methodology

Management Pathway: Standard Phase 1 clinical pharmacology unit observation pathway **Education Protocol (HCP):** Training on protocol-specific SC vs. IV administration techniques and exact PK sampling schedules **Education Protocol (Patient):** Informed consent details and instruction on completing injection site reporting questionnaires **Quality Audit Mechanism:** Bioanalytical sample chain-of-custody tracking, PK assay validation, and protocol deviation monitoring

14. Observation & Follow-Up Schedule

Total Duration: Up to Day 85 (approximately 12 weeks of follow-up) **Onsite Visit Cadence:** Intensive initial inpatient stay followed by frequent outpatient PK sampling visits **Remote Monitoring Frequency:** Standard safety check-ins as defined by unit protocols **Checklist Review Interval:** Real-time during scheduled PK sampling clinic days

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator				[Insert Name]			_____		[DD-MMM-YYYY]							
	Clinical Operations Lead				[Insert Name]			_____		[DD-MMM-YYYY]							
	Lead Statistician				[Insert Name]			_____		[DD-MMM-YYYY]							